Computational and Mathematical Organization Theory (2021) 27:264–301
https://doi.org/10.1007/s10588-021-09331-2

S.I. : SBP-BRIMS2020

Business process management heuristics in IT service management: a case study for incident management

Rúben Pereira[1] · José Braga de Vasconcelos[2,3] · Álvaro Rocha[4] · Isaías Scalabrin Bianchi[5]

Accepted: 23 April 2021 / Published online: 6 May 2021

Abstract

This research aims to understand how Business process management (BPM) can be applied for the improvement of Information Technology service management (ITSM) processes. A case study is conducted for the improvement of the time performance of the incident management process, since it is pointed as a quick win for ITSM. The results obtained identified three best practices—activity automation, activity elimination and integral technology—as the best suited for the improvement of the time performance of the incident management process. Using a simulation tool for business processes, it was revealed that the employment of these best practices in the analysed incident management process eliminates the effort required in the 1st support level and reduces in 10.7% the average processing time in the 2nd support level.

Keywords Information systems · Business process management · IT service management · Case study · Process improvement

✉ José Braga de Vasconcelos
jose.vasconcelos@ulp.pt

Rúben Pereira
ruben.filipe.pereira@iscte-iul.pt

Álvaro Rocha
amr@iseg-ul.pt

Isaías Scalabrin Bianchi
isaias.bianchi@ufsc.br

1 University Institute of Lisbon (ISCTE-IUL), Lisbon, Portugal

2 Lusofona University of Porto (ULP), Porto, Portugal

3 Centre for Public Administration and Public Policies (CAPP), University of Lisbon, Lisbon, Portugal

4 ISEG - Lisbon School of Economics and Management, University of Lisbon, Lisbon, Portugal

5 Universidad de Lima, Lima, Peru

1 Introduction

This paper presents a research approach, applying knowledge engineering techniques for the development of information systems, and focusing on Information Technology Service Management (ITSM) activities. This research also presents an exploratory case study in a multinational company, conducted for the improvement of the time performance of an incident management process.

There are several ITSM frameworks available, consisting in best practices that propose standardizing these processes for the respective operations. By adopting these frameworks, organisations can align information technology (IT) with their business objectives. The constant need for change, competitiveness and innovation in organisations compels managers to analyse its business processes (BPs) and find improvement opportunities. Therefore, the objective of this research is to understand how BPM can be used to improve of ITSM processes. Data were gained through documentation, archival records, interviews, and focus groups with a team involved in IT support service. The results of this study are the as-is process that was elicited and respective incongruences clarified. The preliminary evidence of this research show that may be is possible to improve incident management process using business process modelling. The next steps of this research are to identify the main problems and simulate the appropriate business process management (BPM) heuristics to understand the impact in the business organisation.

Change is a constant in business environments. Classic management authors, such as Porter (1980), helped to understand that change, whether of the internal or of the external environment, affects organisations and that it is an issue that should be addressed. As Harmon (2014, p. 19) argues, "change and relentless competition call for constant innovation and for constant increases in productivity, and those, in turn, call for an even more intense focus on how work gets done. To focus on how the work gets done is to focus on business processes". This means a continuous call for BP change and improvement (Harmon 2014). For this challenge, BPM appears as a discipline for the management of BPs focused on continuous improvement (Dumas et al. 2013). van der Aalst (2013, p. 1) argues that BPM has the "potential for significantly increasing productivity and saving costs" and also states that "BPM has a broader scope: from process automation and process analysis to operations management and the organization of work", providing different ways to approach change and to improve BPs. Best practices have been proposed for improvement initiatives, consisting in redesign heuristics to modify BPs and align them with business objectives (BO)(Hanafizadeh et al. 2009; Mansar and Reijers 2007; Pereira et al. 2020).

With the increasingly pressing developments in communication and technology, IT is seen today as "an integral part and fundamental to support, sustain, and grow a business" (Mohamed et al. 2012, p. 1), being impossible for many organisations to function and succeed without it (Sallé 2004). IT became a complex and dynamic landscape in organisations (Jamous et al. 2017). With this, came the need to align IT operations with the BO, which led gradually to the *servitization* of IT operations (Conger et al. 2008). Thus, IT service management (ITSM) arose. In order to help organisations to perform ITSM, several IT frameworks were proposed, providing managers and organisations with customer-centred sets of processes and best-practices, for managing IT operations and aligning them with their BO (Marrone and Kolbe 2011). As the literature shows (Galup et al. 2009; Iden and Eikebrokk 2013; Pereira and Mira da Silva 2012; Valiente et al. 2012), the most used of these frameworks is Information Technology Infrastructure Library (ITIL). A recent report about the current state of ITSM indicates that 47% of the surveyed IT managers employ ITIL or some of its processes in their ITSM strategy, being the most adopted IT framework (BMC and Forbes Insights 2017).

The constant need for change, competitiveness and innovation in organisations compels managers to analyse its BPs and find improvement opportunities (Harmon 2014). Being BPM a discipline that integrates process-oriented improvement initiatives (Mahy et al. 2016), it is relevant to understand if it can be employed to improve IT services, which are based on processual ITSM frameworks. This research focuses on revealing how BPM can be employed for the improvement of the ITIL's incident management (IM) process, one of the most adopted ITSM processes (AXELOS 2019; Gupta et al. 2008a; Mahy et al. 2016). This is a qualitative and ex post investigation aiming to elicit explore how and which BPM heuristics can be used to improve IM process. Being an underexplored topic, an exploratory case study is conducted, following the guidelines proposed by Yin (2009).

2 Literature review

Business processes have always been part of the organisations, whether in a formal or informal ways (Roeser and Kern 2015). BPs define the tune and performance of an organisation when delivering a service or a product to a customer (Dumas et al. 2013).

Dumas et al. (2013, p. 5), in a 20-year updated definition, establish "business process as a collection of inter-related events, activities and decision points that involve a number of actors and objects, and that collectively lead to an outcome that is of value to at least one customer".

Plus, BPs allow organisations to reach its BOs efficiently and effectively by enabling the coordination and connection of its resources (Weske 2012). Thus, the management and improvement of BPs is vital for organisations, as it allows to achieve BOs while also coping with change (Harmon 2014).

BPM, by implying a continuous commitment to manage BPs, requires a lifecycle methodology with structured steps and feedback that establish a managerial practice in organisations, which is the premise for continuous improvement and to meet the BOs (de Morais et al. 2014). This BPM lifecycle is based in principles such as modelling and documentation of BP, customer-orientation, constant assessment of the performance of BP, a continuous approach to optimization and improvement, following best practices for superior performance, and organisational culture change (Pyon et al. 2011). Several BPM lifecycle proposals appeared, contributing for the growth of BPM as a concept. The BPM lifecycle proposed by Dumas et al. (Dumas et al. 2013), have these phases: process identification, process discovery, process analysis, process redesign, process implementation and process monitoring and controlling. By applying BPM lifecycles, organisations can address BP change and identify the improvements required to achieve the desired BOs (Dumas et al. 2013; Harmon 2010). Recently BPM has even expanded its boundaries being the base of a new domain called Robotic Process Automation (Santos et al. 2019).

Galup et al. (2009, p. 125) state "because ITSM is process-focused, it shares a common theme with the process improvement movement (such as, TQM, Six Sigma, Business Process Management, and CMMI)". This section presents the literature review performed to collect existing related work on the improvement of IM process through BPM. The related work found can be divided in two categories: improvement of the IM process and BP improvement through BPM. Table 1 presents the examples found of work developed on the improvement of the IM process.

The related work found for the improvement of the IM process revealed that most of the work developed is technology-oriented, with different methods and techniques being applied for technological improvement solutions. Automation and incident correlation are the most common improvement methods found. Table 2 presents examples of works developed on BP improvement through BPM.

Several more related works were found concerning BP improvement through BPM, being the most common example located in healthcare services. Although several examples were found in each of the two categories mentioned, there was not a single work found concerning the improvement of IM process through a BPM approach, the topic of this research. Two examples were found that slightly approached this topic:

- The work of Mahy et al. (2016), published in the 3rd International Conference on Systems of Collaboration, that only modelled the IM process.
- The work of Bezerra et al. (2014), published in the 9th International Conference on the Quality of Information and Communications Technology, that also only modelled the IM process.

Based on the previous paragraphs one may note that no other research explores the relation between BPM heuristics in IM process. Therefore, there is a gap that this investigation intends to fulfil by answering the following research question: "What are the BPM redesign heuristics that best suit IM process improvement?".

3 Research methodology

Grounded on the previous section, one may argue that this study is exploratory in nature rather than hypothesis testing. Exploratory analysis should be considered where there are none or few prior works presented on the subject studied (Yin 2009; Zaidah 2007). Reinforced by Runeson and Höst (2009) as having exploration as primary objective, to Zainal (2007) a case study analysis is suitable to analyse a limited

Table 1 Related work on the improvement of the IM process

Authors	Publication	Improvement method
Ghrab et al. (2016)	3rd International Conference on Artificial Intelligence and Pattern Recognition	Automation through constraint programming
Goby et al. (2016)	24th European Conference on Information Systems	Automation through business intelligence
Salah et al. (2015)	Journal of Network and Systems Management	Incident correlation model
Bezerra et al. (2014)	9th International Conference on the Quality of Information and Communications Technology	Optimization through the reuse of experiences and natural language processing techniques
Bartolini et al. (2008)	19th IFIP/IEEE International Workshop on Distributed Systems	Optimization through discrete event simulator
Gupta et al. (2008b)	5th IEEE International Conference on Autonomic Computing	Automation through incident correlation

Table 2 Related work on the BP improvement through BPM

Authors	Publication	Area
Sanka Laar and Seymour (2017)	13th European Conference on Management, Leadership & Governance	Small and medium enterprises
Rebuge and Ferreira (2012)	Journal of Information Systems	Healthcare services
Netjes et al. (2010)	Workshops of 7th International Conference on Business Process Management	Healthcare services
Becker et al. (2007)	13th Americas Conference on Information Systems	Healthcare services
Küng and Hagen (2007)	Business Process Management Journal	Banking services
Hertz et al. (2001)	Supply Chain Management: An International Journal	Automotive industry
Ferretti and Schiavone (2016)	Business Process Management Journal	Operations in seaports
Rinaldi et al.(2015)	Business Process Management Journal	Public administration
Haddad et al.(2016)	Business Process Management Journal	Non-profit organisations

number of events or conditions in detail. Thus, this research follows the case study methodology proposed by Yin (2009).

Following Yin (2009) recommendations, a research question is formulated, to fulfil the purpose of this study. What are the BPM redesign heuristics that best suit IM process improvement? This question will guide the authors in the research for a common ground between BPM and an ITSM process that has not been proper explored.

3.1 Case study design

Table 3 presents the case study stages performed according to guidelines proposed by Yin. Although the author defines these stages as part of a logical case study process, he also argues that this process is iterative and has a certain degree of flexibility, allowing repetitions and parallelisms in its execution.

To assure the quality of the case study results, the four validity tests of [48] are pursued throughout the research. The validity of the case study will be analysed in Sect. 8.

To support the conduction of the case study, a specific BPM system will be used as tool for modelling, analysis and simulation purposes. The language employed for modelling is the Business Process Model and Notation 2.0 since it is a recent and broad language that combines other modelling best practices, which makes it easier to interpret and understand.

4 Research context unit of analysis

The unit of analysis of this case study will consist in a team that belongs to a multinational company in the markets of electrification, digitalization, and automation, which is present in at least 190 countries and employs directly more than 350.000 people. The team was formed in 2014 and is composed by 17 members, being characterized in Table 4.

Table 3 Case study stages

Stages	Description	Sections
Plan	Identify case study as the adopted methodology and define research questions	1, 2 and 3
Design	Identify the specific case study design and establish its logic and quality procedures	3.1 and 3.2
Prepare	Define the protocols and develop the skills to collect case study evidence	3.3
Collect	Follow the protocols and collect evidence from different sources, building and maintaining a chain of evidence	4
Analyse	Conduct the analysis and interpretation of the collected evidence	5
Share	Develop conclusions related to data analysis results and compose a case study reporting	6

Adapted from Yin

Table 4 Team members information

ID	ITIL experience	Time (in years)		Function
		IT experience	With IM team	
I1	Yes	19	3	Team leader
I2	Yes	16	3	Support expert
I3	No	0,16	0,16	Support expert
I4	No	5	2	Support expert
I5	Yes	7	3	IT specialist
I6	Yes	23	2	Support agent
I7	Yes	22	3	Support expert
I8	No	3	0,16	Support expert
I9	Yes	22	3	IT specialist
I10	Yes	5	2	IT specialist
I11	Yes	13	2	Support expert
I12	Yes	4	2	Support expert
I13	Yes	25	3	Support expert
I14	Yes	14	3	Support expert
I15	Yes	9	1	Support expert
I16	No	0,25	0,25	Support agent
I17	Yes	18	3	IT specialist
	Average	12,1	2,1	

The team is based in Lisbon, Portugal, integrated in an IT support service to the whole Human Resources (HR) department of the company, being the owner of a local IM process. With this process, the organisation provides an IT support service that receives and processes all incidents reported by the users of the HR's IT services, which are the customers of the process. This team is chosen as the unit of analysis for this case study because it is the owner of an IM process, the selected ITSM process for research. Being the goal of the team and of its IM process to provide a fast service, time is considered as the main performance dimension. Thus, time shall be the driving performance dimension for the improvement proposals in this research.

4.1 Data collection

A plan for collection of evidence was defined following Yin (2009) recommendations. To achieve the validity of the analysis that will be performed and avoid the weaknesses inherent to each source of evidence, it is desirable to collect evidence from difference sources. For this case study, four different sources of evidence are expected to be made available by the organization, as presented in Table 5.

Being an IT-nature process, observation and physical artefacts are not expected to be available as sources of evidence. Adding to the enounced sources, it is also expected that informal sources, such as punctual conversations, may occur with

Table 5 Expected sources of evidence

Sources of evidence	Description
Documentation	Internal documentation and private web content about the organisation, the IT support service, the team, and the IM process
Archival records	Data records and data reports generated by the daily operation of the IM process
Interviews	Open and focused interviews, with all the team members. Four rounds of interviews
Focus groups	Structured focus groups, with all the team members

team members. The plan for collection of evidence is designed according with the different objectives of each BPM lifecycle phase (Dumas et al. 2013) and also taking into account that the performance dimension to improve in this research is time.

5 Process discovery and analysis

The goal of Process Discovery is to document the current state of the process, by producing an as-is model (Dumas et al. 2013). The process discovery steps can be seen in Fig. 1. The first step consisted in the collection of the evidence required, through the conduction of the two rounds of interviews and gathering of documentation. The analysis of the evidence was performed in parallel with its collection. This analysis provided the desired information for this phase and allowed to clear the incongruencies detected in the collection of evidence. The IM process was then documented in its current state, through an as-is model and respective description. Lastly, the 1st focus group was convened to approve the documentation.

The 1st round of interviews followed a script directed to the mapping of the process. The interviewees were asked to map the IM process, from end–to–end, detailing the activities performed, the participants involved, the decisions and exceptions along the process.

The 2nd round of interviews was conducted based on the results of the previous round of interviews. This time, the team members were faced with the initial

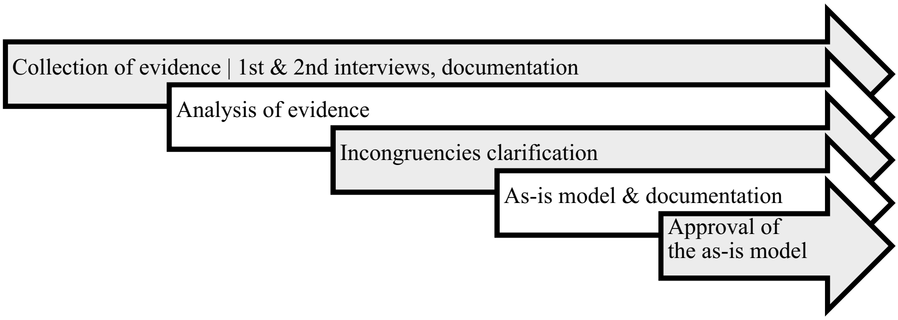
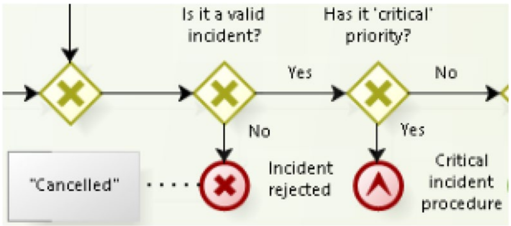
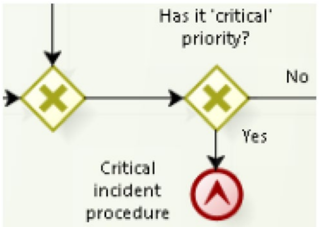


Fig. 1 Process discovery steps

Table 6 Incongruency A

Incongruency A	
Scenario 1	Scenario 2
	

Table 7 Incongruency B

Incongruency B	
Scenario 1	Scenario 2
	

draft of the as-is model and requested to validate it: to spot and correct inaccuracies, and to address incongruencies and clarify them. They were also asked to detail even more the model with relevant artefacts and information inherent to the IM process.

Documentation was collected, being available administrative documents about the support service, descriptions of the participants in the IM process and respective roles, as well as a depiction of the workflow adopted.

To have an approved as-is model, the 1st focus group was convened. The team was asked to collectively analyse the final as-is model, discuss its validity and point any required adjustments. These methods of evidence collection were employed because they enable a thorough Process Discovery, having as output the documentation of the as-is model, which is the basis for the analysis of the IM process and its issues.

In the two rounds of interviews conducted, four incongruencies were detected, concerning the mapping of the IM process.

Incongruency A, depicted in Table 6, had its source in the 1st round of interviews, and concerned what of activity the Incident Prioritization in the 1st support level is: a user activity, performed by a team member (scenario 1), or a script activity, automatically performed the IM system (scenario 2).

The 2nd round of interviews cleared this incongruency, by defining Incident Prioritization has a script activity performed automatically by the IM system (scenario 2).

Table 8 Incongruency C

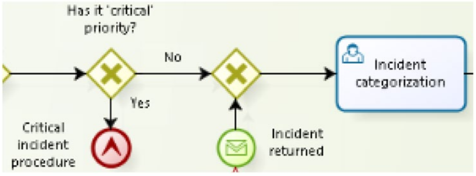
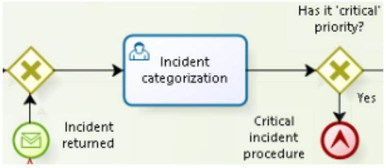
Incongruency C	
Scenario 1	Scenario 2
	

Table 9 Incongruency D

Incongruency D	
Scenario 1	Scenario 2
	

Incongruency B, shown in Table 7, had its source in the 1st round of interviews, and concerned existence of a gateway for the triage of incident validity in the 1st support level (scenario 1) or not (scenario 2).

The documentation analysis solved this incongruency, by revealing that there was no gateway for the triage of incident validity in the 1st support level (scenario 2).

Incongruency C, presented in Table 8, also had its source in the 1st round of interviews, and concerned the position of a gateway for the triage of critical priority incidents in the 1st support level: if before the Incident Categorization activity (scenario 1) or after (scenario 2).

The 2nd round of interviews cleared this incongruency, by setting the position of the gateway before Incident Categorization activity (scenario 1).

Incongruency 4, presented in Table 9, was detected in the 2nd round of interviews, and concerned the period of working days required by the IM system for closing an incident: if 14 working days (scenario 1) or 7 working days (scenario 2).

The documentation solved this incongruency, by revealing 14 working days as the time period required by the IM system for closing an incident (scenario 1). With all the incongruencies cleared, the final documentation of the IM process was produced, including the final draft of the as-is model.

There are three main identified participants in the IM process:

- The customers, which are the users who report incidents to the support service.
- The support service, composed by the three support levels (SL) that perform the IM process itself, thus being the focus of this research.

Table 10 Support levels in the support service

Support level	Description & responsibilities	Staff
1st SL	*Support agents that perform the initial reception, triage and forwarding of incidents to the 2nd SL, the so-called dispatching activities*	*2*
2nd SL	*Support experts that perform a technical diagnosis and resolution of the incidents, being also responsible to contact the customer and to always close incidents. If unable to find the solution for the incidents, it must forward it for the 3rd SL*	*10*
3rd SL	*IT specialists that perform an extensive investigation and resolution, being the last resort of the support service to solve the incidents. If unable, it must request for the intervention of the respective IT supplier. With the resolution performed, the3rd SL must return the incident to the 2nd SL for closure*	*4 (in the team)*

- The IT suppliers, which are external to the support service and are the providers the IT services used by the organisation.

The support service is composed by three SLs that have different roles and teams involved, as presented in Table 10.

Of the three SL presented in Table 10, the team fully incorporates the 1st SL with 2 support agents, and the 2nd SL with 10 support experts. However, the 3rd SL is partially represented in the team, with only 4 IT specialists. The 3rd SL is composed by several IT specialists from various IT development teams in the organisation, that are called to participate in the IM process whenever required, being the quantity of staff involved in this SL unknown. This structure of the support service is designed to handle and solve the incidents according with their complexity and severity, and with the level of expertise and specialization required, being one of goals of the team to retain and solve as much incidents as possible in the 2nd SL, avoiding a high workload for the 3rd SL.

The IM process starts whenever a customer reports an incident to the 1st SL, either through the email (Incident reported) or through call (Call arrival). From here, the IM process is mainly grounded in the workflow defined by ITIL, being its activities and gateways easily recognized in the as-is model. These activities are performed with the support of a single IM system, which is used for the storage and management of all incidents, through the logging and update of incidents in tickets with all the respective information and actions performed.

There are two rework situations in the middle of the IM process:

- Whenever there is an incident returned from the 2nd SL to the 1st SL, due to mistake of the 1st SL.
- Whenever there is a reopening request from the customer to the 2nd SL, after an incident is labelled as resolved.

The IM process finishes in the Incident solved & service restored, the main end event of the process in the 2nd SL, with the incident solved and the ticket closed. However, two alternative end events may occur:

- Incident rejected, when the 2nd SL determines that there is no valid nature in the received incident that justifies the deployment of the IM process and cancels the respective ticket.
- Critical incident procedure, a special procedure designed to deal with incidents that have a perceived critical priority or critical nature. These procedures are different and customized according with the different IT services.

The documentation of the IM process reveals that, despite being adapted to the organisation, it is mainly grounded on the ITIL standard and that most of the recommendations proposed by the standard are followed. The final documentation of the IM process was submitted for approval in the 1st focus group. This 1st focus group, convened with all the team members, validated the documentation,

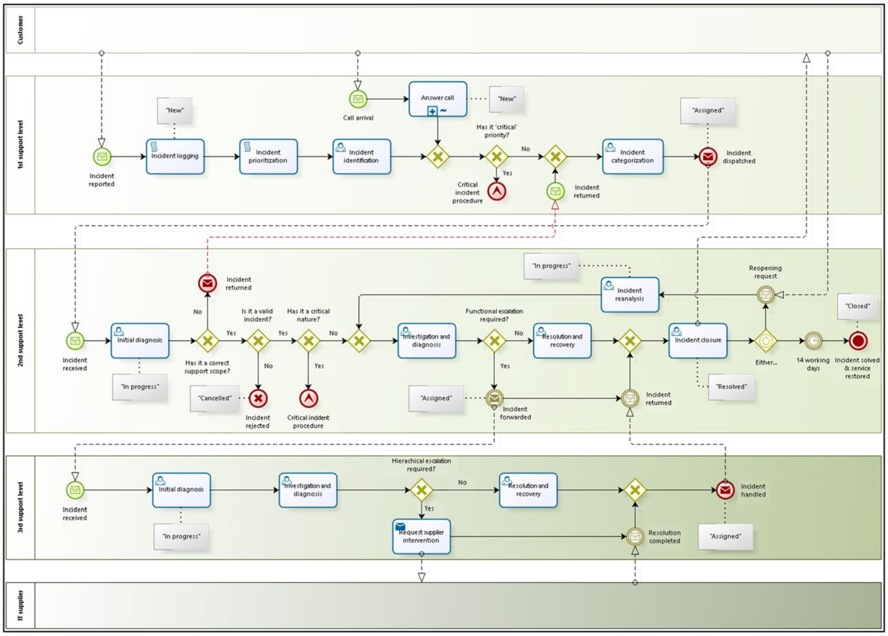


Fig. 2 AS-IS model

without any opposition, not being required any adjustments or corrections to the as-is model. The AS-Is model can be seen at Fig. 2.

In Process Analysis, the issues affecting the process in its current state are identified and quantified using a performance dimension, being the output a structured collection of issues. Considering time as the main driver for this research, the Process Analysis was focused on unravelling the existing issues affecting the time performance of the IM process. The steps followed in this stage are listed in Fig. 3.

In the 3rd round of interviews, the team members reported and described the main problems affecting the IM process and challenges to the daily operation. These various issues were compiled and listed, as presented in Table 11.

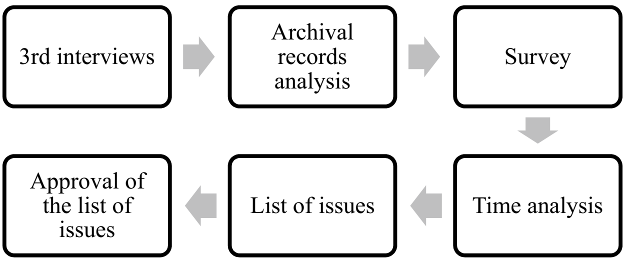


Fig. 3 Process analysis steps

Table 11 Issues reported by the team members

ID	Reported issues
A	3 interviewees (I6, I8 and I16) reported that answering calls in the 1st SL to receive incidents is a too much time-consuming activity, which takes time that can be applied in the remainder dispatching activities
B	10 interviewees (I1, I2, I4, I7, I11, I12, I13, I14, I15 and I17) mentioned that sometimes the dispatching performed by the 1st SL is done wrongly, which leads to a waste of time and effort by the 2nd SL on incidents with incorrect scope
C	9 interviewees (I1, I2, I4, I5, I7, I8, I12, I16 and I17) implied that the dispatching activities of the 1st SL are time-consuming and prone to errors, being referred also as essential targets for analysis and improvement
D	All the interviewees indicated that the IM system utilized by the support service is outdated, with time-consuming features, low automation degree and lacking key reporting tools

Being time the driving dimension of this research, it is clear, through the identification of the issues A, B and C, that the 1st SL must be a focal point of further analysis regarding the its time performance and how it affects the overall IM process. Issue D suggests that some of the features in the IM system are inadequate to the daily operation and can affect the IM process time performance. Despite this, issue D is not related to a measurable activity, which limits the conduction of a time analysis on it. However, this issue may be reflected in the duration of the activities of the IM process. Also, the fact of not being measurable does not mean that improvement suggestions cannot be proposed.

The analysis of the archival records was performed on the 6385 incidents collected from the IM system. Performing a simple Pareto analysis, it was found that 9 IT services (~20% of the IT services) corresponded to approximately 87% of the whole demand, as shown in Table 12.

This reveals that there is a great inequality in the distribution of the demand per IT service, probably due to the nature of the various IT services. The demand

Table 12 Pareto analysis of the IM process demand

IT services	Volume (q)	Weight rate (%)
A	1790	28.03
B	1064	16.66
C	554	8.68
D	528	8.27
E	500	7.83
F	344	5.39
G	294	4.60
H	259	4.06
I	219	3.43
9	5552	86.95

Table 13 Arrival rates of the IM process

Starting events	Volume	Rate (%)	Arrival rate (m)
Incident reported	6025	94.36	13.862
Call arrival	360	5.64	232.00
	6385	100	

is polarized to a distinguished set of IT services to which the IM process is more requested for.

It was found that, out of the 6385 incidents collected, 360 incidents were reported through calls, while the remaining 6025 incidents were reported through email. Based on this, there were a total 58 working days, the arrival rates of the starting events were calculated in minutes (m), as presented in Table 13.

Table 13 reveals that the most frequent way for customers to contact the support service is through email, being calls a small rate in the processed volume. These conclusions are relevant for the quantification and analysis of issues A and C.

Rates concerning the handling of incidents between each participant—incident returned (from the 2nd SL to the 1st SL), incident forwarded (from the 2nd SL to the 3rd SL) and supplier intervention requested—were obtained through archival records concerning the self-solving quotas of each SL of the support service. Although there were no volumes associated with these rates, considering that the processed volume was 6385 incidents, such quantities can be estimated. Table 14 shows those rates (%) and the respective estimated volumes (q).

In the case of supplier intervention requested (*) the rate refers to the volume that arrived to the 3rd SL and had to be forwarded to the IT supplier. Therefore, the calculation of the respective quantity is made with the estimated volume of the 3rd SL (2171 incidents) and not with the overall processed volume (6385 incidents). The rate and estimated volume for incident returned (from the 2nd SL to the 1st SL) are relevant for the analysis of issues B and C, since it reflects the impact of the 1st SL and of dispatching errors in the IM process.

Table 14 Estimated quantities for handling rates

Events/activities	Rate (%)	Estimated volume (q)
Incident returned (from the 2nd SL to the 1st SL)	12	766
Incident forwarded (from the 2nd SL to the 3rd SL)	34	2171
Supplier intervention requested	5	109

Table 15 Estimated quantities for the opposite handling rates

Opposite events/activities	Opposite rate (%)	Estimated volume (q)
No *incident returned* (from the 2nd SL to the 1st SL)	88	5619
No *incident forwarded* (from the 2nd SL to the 3rd SL)	66	4214
No *supplier intervention requested*	95	2062

Also, important to know are the opposite volumes (q) and rates (%) for those handling rates, as Table 15 presents.

The values in Table 15 show that 5619 incidents were correctly dispatched by the 1st SL, that 4214 incidents were solved entirely by the 2nd SL, and that 2062 incidents in the 3rd SL did not require the intervention of an IT supplier.

The processing volumes (q) and rates (%) for the remainder events were also provided by the archival records, as presented in Table 16.

All these volumes and rates enabled to characterize some of the flows of the IM process:

- The most probable path, also the desired one by the support service, consists in an incident arriving through email, being correctly dispatched from the 1st SL, being solved entirely by the 2nd SL and not having a reopening request.
- The second most probable path consists in an incident arriving through email, being correctly dispatched from the 1st SL, solved by the 3rd SL, closed by the 2nd SL, without a reopening request.
- The least probable paths concern the incidents that finish in critical incident procedure and in incident rejected.

The analysis to the KPIs and SLAs compliance rates allowed to assess if the IM process was meeting its objectives. The KPIs and SLAs used by the team were analysed:

- The team has as KPI the self-solving quota for the 2nd SL, since the 2nd SL is fully incorporated in the team. The target established for this KPI defines 60% as the minimum rate for self-solving quota in the 2nd SL. As seen above in Table 15, the team is meeting this objective by having reported 66% rate of self-solving quota in the 2nd SL (the opposite of the 34% of incidents forwarded to the 3rd SL).
- The team also uses the rate of reopening requests as KPI. The target established for this KPI sets 5% as the maximum rate allowance. As presented above in Table 16, the team is meeting this target by having only a 2.99% rate of reopening requests.
- The team has time effort defined as KPI for time performance. This KPI measures the time of effective work for processing incidents in each SL, thus concerning the processing times. However, it was discovered that the IM system was not recording this parameter, which meant that there were no processing

Table 16 Processing rates in the incident management process

Events	Volume (q)	Rate (%)
Critical incident procedure (overall)	8	0.13
Incident rejected	99	1.55
Reopening request	191	2.99

times available. It also meant that the team had no perspective or control on what the time performance of the IM process was, which is equal to say that there were no insights on how the work was being performed. The lack of processing times in the archival records limited the conduction of the planned time analysis. This limitation was also relevant for issue D, by being related with the obsolescence of the IM system.

- The SLAs for this IM process are time limits defined for the resolution of incidents, which vary depending on the priority and classification of the incidents. By analysing the SLAs compliance rates, it was found that no single incident of the collected 6385 incidents exceeded the imposed the time limits, which allows to conclude that the IM process is assuring the quality of the service it provides.

The existence of processing times records was necessary to perform the planned time analysis. To overcome the inexistence of such records, a survey was conducted with all the team members to collect estimates for the time required to accomplish each user activities of the IM process, as described in previous section. The output of this survey consisted in processing time estimates, in minutes (m), for each activity, as shown in Table 17.

For the script activities of incident logging and incident prioritization, which are automatically performed by the IM system, no estimates were requested, since they do not require effort from the 1st SL to be accomplished. So, these activities are considered to have null processing times.

With the time estimates for the IM process activities collected, it became possible to quantify and configure the AS-IS model, in order to perform the time analysis.

The time analysis was performed through the simulation tool on the BPM system used to support this research. With the as-is model quantified and configured with the data previously collected, the simulation of the IM process in its current state was run with 50 replications, to assure the stability and reliability of the results.

The outputs of this time analysis were the average, maximum and minimum times required for processing incidents in each SL and also for the execution of each activity of the IM process. This analysis allowed to understand the overall processing time required by the support service to perform the IM process and solve incidents.

Table 18 exhibits the average processing time, in minutes (m) and hours (h), in each SL.

This demonstrates that incidents usually take a combined 3.54 h in 1st SL and 2nd SL. However, the 34% of incidents that are forwarded to the 3rd SL require in average almost 12 h of processing. Although these values provide an insight of the processing times required for each SL, they are averaged values, which means they do not reflect the reality of different, more extreme processing times. These processing times may vary depending on the incident nature and on the various paths it can follow in the IM process. Therefore, the processing time for each incident is always different. Table 19 shows the average time required, in minutes (m) and hours (h), for processing incidents in the two most common paths.

The values in Table 19 demonstrate a big difference between the two most probable paths. The most probable path requires almost 3 times less time than the

Table 17 Time estimates for the 1st/2nd/3rd support level activities

Activities	Dimension	Time (m)
1st support level time estimates		
Answer call	Maximum	45
	Most Likely	16
	Minimum	10
Incident identification	Maximum	8
	Minimum	2
Incident categorization	Maximum	5
	Minimum	1
2nd support level time estimates		
Initial diagnosis	Maximum	18
	Minimum	5
Investigation and diagnosis	Maximum	145
	Most Likely	16
	Minimum	6
Resolution and recovery	Maximum	559
	Most Likely	18
	Minimum	7
Incident closure	Maximum	11
	Minimum	3
Incident reanalysis	Maximum	20
	Minimum	8
3rd support level time estimates		
Initial diagnosis	Maximum	23
	Minimum	5
Investigation and diagnosis	Maximum	1000
	Most Likely	42
	Minimum	4
Resolution and recovery	Maximum	1020
	Most Likely	50
	Minimum	7
Request supplier intervention	Maximum	5
	Minimum	1

second most probable path. Again, these are averaged values, which means that

Table 18 Average processing time for each support level

Pool	Average processing time	
	(m)	(h)
1st support level	9.45	0.16
2nd support level	202.69	3.38
3rd support level	707.77	11.80

Table 19 Average processing time in the two most probable paths

Paths	Average processing time	
	(m)	(h)
Most probable	276.06	4.60
Second most probable	803.02	13.38

they do not reflect more extreme processing times required for both paths.

The time analysis also enabled to address and quantify some of the issues identified in the 3rd round of focused interviews.

To quantify issue A, an analysis was performed to determine the average time spent per call and the total time spent on calls, which are presented, in minutes (m) and hours (h) in Table 20.

These results revealed that the 1st SL spent a total of 141.75 h (approximately 6 days) answering calls to receive incidents. Considering that calls only represent 5.64% of the whole processed volume, this proves that answering calls is a time-consuming activity.

The analysis to quantify issue B aimed at revealing the time that 1st SL and 2nd employed in dealing with dispatching errors. Table 21 shows the results in minutes (m) and hours (h).

This analysis discovered that the 2nd SL spent 146.78 h dealing with dispatching errors, while the 1st SL spent 38.28 h, a total combined of 185.06 h. This means that each of the 10 support experts in the 2nd SL dedicated 14.68 h to deal with errors incoming from the 1st SL, and that each of the 2 support agents in the 1st SL had to rework 19.14 h to correct those categorization mistakes.

Table 20 Issue A analysis

Issue A: Time spent with calls		
Impact to the 1st support level		
Volume	(q)	360
Average time spent per call	(m)	23.65
Total time spent in calls	(m)	8504.81
	(h)	141.75

Table 21 Issue B analysis

Issue B: Time spent with dispatching errors				
Impact		In the 1st support level	In the 2nd support level	Total
Volume	(q)	766 dispatching errors		
Average time spent per dispatching error	(m)	3.00	11.50	14.50
Total time spent with dispatching errors	(m)	2296.97	8806.56	11,103.54
	(h)	38.28	146.78	185.06

Table 22 Issue C analysis

Issue C: Time spent with 1st support level activities			
Activities/pool	Average time (m)	Maximum time (m)	Minimum time (m)
Answer call	23.65	43.38	10.68
Incident identification	5.00	8.00	2.00
Incident categorization	3.00	5.00	1.00
1st support level	9.45	47.03	1.00

Table 23 Issue E analysis

Issue E: Time spent with incidents rejected				
Impact		In the 1st support level	In the 2nd support level	Total
Volume	(q)	99 incidents rejected		
Average time spent per incident rejected	(m)	3.00	11.50	14.50
Total time spent with incident rejected	(m)	296.87	1138.19	1435.05
	(h)	4.95	18.97	23.92

Dispatching errors are clearly a waste of time in the IM process, not only because they require rework and time spending but also because they are not supposed to happen.

Issue C was addressed by extracting the average, maximum and minimum processing time for the 1st SL and its activities, which are detailed, in minutes (m), in Table 22.

The minimum processing time in 1st SL is 1 min, which is related to the dispatching errors that return to the 1st SL for rework and correction. The average processing time in the 1st SL is 9.45 min, which is long for a set of activities that should be performed swiftly, almost automatically. For the worst-case scenario, the results show that the maximum processing time may be 47.03 min, which is almost 5 times more than the average processing time. The 1st SL activities should be performed quickly but are not. This is due mainly to error and slowness of the HR, but also due to calls, proven once again as time-consuming.

While analysing issue B, another related issue was perceived and unravelled: the time spent by the 1st SL and the 2nd SL with incidents rejected. Table 23 presents the new identified issue E, in minutes (m) and hours (h).

Analysis of issue E revealed that a combined total of 23.92 h (nearly one day) were spent with incidents that were eventually rejected, which is high for a very low volume event. Although the rejection of invalid incidents is natural to the IM process, in this

Table 24 Results of the *Process Analysis* steps

Steps	Results
3rd interviews	The identification and description of 4 issues (A, B, C and D) that affect the time performance of the IM process, as reported by the interviewees
Archival records analysis	Allowed to characterize the demand and assess on the compliance of the IM process. Provided insights on the IM process, including processing rates, which are essential to perform the time analysis
Survey	Provided estimates for the processing times of all activities in the IM process, which is required to perform the time analysis
Time analysis	Revealed the time performance of the IM process and quantified issues A, B and C, allowing also to identify and quantify issue E, revealing the impact of these issues in the IM process

Table 25 Contributions of the *Process Analysis* methods for the 5 issues

Contributions	3rd interviews	Archival records analysis	Survey	Time analysis
Qualitative insights	Issues A, B, C, D	Issues A, B, C, D	–	Issues A, B, C, E
Quantitative insights	–	Issues A, B, C	Issues A, B, C	Issues A, B, C, E

case, it implies the waste of an average of 14.50 min until the incident is rejected. This processing time is unnecessarily high to a triage that could imply less time waste.

The time analysis quantified issues A, B and C, and also allowed to identify and quantify issue E, describing the impact of these issues in the IM process time performance.

The main results of each Process Analysis steps are indicated in Table 24.

Table 25 shows the contributions of each step for the understanding of the five identified issues.

With all the issues gathered, the final list of issues was elaborated, as presented in Table 26.

The final list of issues was submitted for debate and approval in the 2nd focus group. The 2nd focus group, which gathered all the team members again, debated the five identified issues and their impacts in the IM process. The final list was approved in consensus, with no vote against. Remarks must be made on the reactions of some team members that did not expected such results concerning the quantified issues. Also, suggestions were made to study automation as a possible solution for the issues.

This 2nd focus group marked the conclusion of Process Analysis.

Table 26 Final list of issues

ID	Issues	Short conclusion
A	Calls are too much time-consuming	141.75 h were spent on answering calls in the 1st SL, which represents only 5.64% of the processed volume
B	Dispatching errors are time-consuming	185.06 h spent along the 1st SL and 2nd SL to correct errors that should not happen
C	1st SL activities are time-consuming	Has a high average processing time of 9.45 min, with a worst-case scenario of 47.03 min
D	IM system is obsolete	Lacks KPIs parameters that difficult the management of the IM process, and has old features that are no longer required
E	Invalid incidents are time-consuming	23.92 h were spent to deal with incidents that were rejected

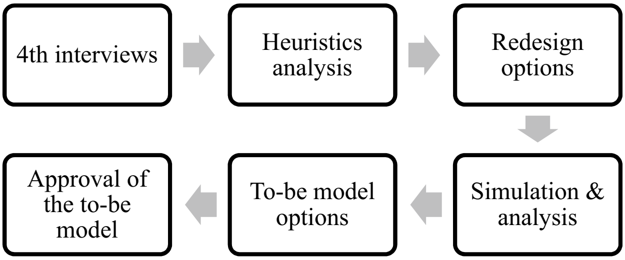


Fig. 4 Process redesign steps

6 Process redesign

Process Redesign consists in identifying and analysing changes to the process that address the issues previously identified, in order to improve the process performance, being the output of this phase a to-be. To achieve this, several steps were taken, as depicted in Fig. 4.

The first step was the conduction of the 4th round of interviews, which collected improvement proposals suggested by the team members and provided hints on how to redesign the IM process.

Secondly, the redesign heuristics proposed in the framework of Becker et al. (2007) were analysed, to identify and select the most adjusted redesign heuristics for the improvement of the IM process. The criteria followed to select the heuristics consisted in:

1. Reject all the redesign heuristics that had a neutral or negative effect in the time performance dimension, according with Becker et al. (2007).
2. Select the redesign heuristics best suited to address the existing issues of the IM process and the improvement proposals, considering also the compliance with the ITIL standard.

With the heuristics identified, different redesign options for the IM process were produced by employing and combining those heuristics. Each redesign option was then simulated in the BPM system, to measure the respective impact in the IM process time performance. The results of all the different redesign options were compared. The redesign options that presented the best time performance improvements were selected as possible to-be models and presented in the 3rd focus group for debate. The 3rd focus group was convened to debate the best redesign options and identify one as the to-be model for a new, improved IM process.

The established plan for evidence collection was followed in this phase. However, in this phase, due to organisational change, team member I17 was no longer part of the team. Therefore, it did not participate in the evidence collection moments of this phase involving the team members. Table 27 presents the details of the employed methods.

Table 27 Details of evidence collection in *Process Redesign*

Method	Date/Period	Approximate duration	Participants	Annexe
4th interviews	May 7th to May 18th	30 min each	All team members, individually, except team member I17	6
3rd focus group	June 26th	1-h	All team members, collectively, except team member I17	–

The 4th round of interviews followed a script aimed at its goal of gathering redesign and improvement proposals suggested by the team. The as-is model and the list of issues from the previous phases were presented to the team members, which were given time to think and requested to suggest, though not forced, any proposals for redesign and improvement.

To close this phase, after the analysis of the possible redesign options, a final 3rd focus group was convened to decide on the final to-be model. All redesign options, including the ideal redesign option, were presented, along with the issues that they would solve. The team members were then asked to debate on each redesign option and to choose which one is the most suitable for implementation, presenting the reasons why. This allowed to choose and approve the best suited to-be model for this IM process.

Despite being a more analysis-oriented phase, the moments of evidence collection gave sustenance and reason to the analysis realized and to the decisions made regarding the redesign options and the final to-be model. IT also answered the research question of this case study.

Table 28 describes the outputs of two moments of evidence collection occurred in this phase.

In the 4th round of interviews, the team members suggested redesign and improvement proposals, based on the issued identified in Analysis. Table 29 lists and describes all the proposals made.

Only a total of three improvement proposals were produced, with neither addressing issue E. These proposals were considered for the selection of the best suited redesign heuristics.

Of the 29 redesign heuristics listed by Becker et al. (2007), 8 heuristics were initially cast out due to having neutral or negative effects in the time performance. This reduced the number of possible redesign heuristics to 21, all with positive effects on the time performance.

Table 28 Outputs of the collection of evidence in *Process Redesign*

Method	Output
4th interviews	A list of redesign and improvement proposals, suggested by the team members
3rd focus group	Discussion of the redesign options and approval of the final to-be model, by all the team members

Table 29 Improvement proposals collected

ID	Improvement proposal	Issue addressed
P1	6 interviewees (I1, I2, I6, I8, I2 and I16) suggested to end calls in the 1st SL, establishing email as the only way to receive incidents	Issues A, C
P2	11 interviewees (I1, I2, I4, I5, I7, I8, I11, I12, I13, I14 and I15) suggested to automate all the activities of the 1st SL, by integrating them in the IM system, implying also the end of calls as an entry point and as an activity	Issues A, B, C
P3	All the interviewees mentioned that the IM system should be updated or changed to a more recent version that includes time-saving features and automation tools, being the latter suggested as a complement of *P2*	Issues A, B, C, D

Table 30 Identified heuristics for *Process Redesign*

ID	Redesign heuristic	Targeted elements of the current IM process	Issue addressed	Proposal addressed
H1	Activity elimination	*Call arrival* & *answer calls*	Issues A, C	P1
H2	Resequencing	*Initial diagnosis* (in the 2nd SL) and its following gateways	Issues B, E	-
H3	Activity automation	All user activities in the 1st SL	Issues A, B, C, D, E	P2, P3
H4	Integral technology	IM system	Issues A, B, C, D, E	P2, P3

Table 31 Expected effects of the selected heuristics

ID	Expected effect in the IM process time performance
H1	Reduction of the average processing time for the 1st SL
H2	Reduction of the average processing time for the 2nd SL
H3	Elimination of all time spent in the 1st SL activities
H4	Indirect: elimination of all time spent in the 1st SL activities

To these 21 heuristics, the analysis sought to identify which ones best corresponded to the identified issues and to the improvement proposals, based on their nature. Issues A, B, C and E were clearly operational issues, concerning activities and flows of the IM process, while issue D had a technological nature. As for improvement proposals, P1 offered an operational solution to the operational issues A and C. However, P2 and P3 suggested technological solutions for issues A, B, C and D. With this the analysis was narrowed to three categories of redesign heuristics: BP operation heuristics, BP behaviour heuristics and technology heuristics.

To these three categories of redesign heuristics, the analysis aimed at selecting the ones that best suited the improvement of the time performance of the IM process, while still meeting the ITIL standard. For redesign heuristics were selected, as detailed in Table 30.

These four heuristics were selected because they addressed not only the identified issues in the IM process and the improvement proposals, but also because they respect the ITIL recommendations. The expected effect of each heuristic in the time performance of the IM process is described in Table 31.

The premise to employ these heuristics is to combine them:

- H2 should be combined with H1, otherwise the solo employment of H2 will increase an already high average processing time for the 1st SL, since calls would still be performed.
- H3 must be employed with the H4. Despite the latter not having a direct measurable impact in the time performance, it enables for H3 to be employed. Thus, as shown previously, the impact of H3 is the indirect impact of the H4.

Table 32 Redesign options produced

ID	Heuristic employed	Issue affected	Corresponding proposal	Annexe
R1	H1	A, C	P1	7
R2	H1, H2	A, B, C, E	–	8
R3	H1, H3, H4	A, B, C, D	P2, P3	9
R4	H1, H2, H3, H4	A, B, C, D, E	–	10

- H3 must also be employed with H1, or else there will not be a single-entry point of incidents, which is required for centralizing the incoming flow of incidents in one single 1st SL path and automate of the respective activities.

Concluding, the combination of these four redesign heuristics is essential to obtain the best time performance improvement for the IM process.

Based on the identified heuristics and the conditions for their employment, four different redesign options for the IM process were produced, as detailed in Table 32.

Option R1 is the solo employment of H1, proposing the elimination of the time-consuming calls in the 1st SL and centralizes emails as the single-entry point for incidents in the IM process. Option R2, adding to H1, proposes the transfer of the Initial diagnosis activity from the 2nd SL to the 1st SL. Option R3, adding to H1, proposes the automation of the all the user activities in the 1st SL (incident identification, incident categorization) by upgrading the IM system. Option R4, adding to H1, combines the automation of all the user activities in the 1st SL through the upgrading of the IM system with the transfer of the Initial diagnosis activity from the 2nd SL to the 1st SL. Two outcomes, transversal to all the redesign options, are expected:

- Changes in the processing times for 1st SL and 2nd SL.
- No changes in the processing time for the 3rd SL, since neither identified issue or redesign options changes affects it.

The four redesign options were configured for simulation in the BPM system, to measure the time performance of each. The configuration and quantification of each redesign option was based on the data used in the Process Analysis.

All the redesign options imply the elimination of calls and centralization of emails as the single-entry point of the IM process. This means that, with only one starting event, the volume that corresponded to calls (360 incidents) shifted to that entry point.

Therefore, the whole volume of 6385 incidents arrived at the IM process by email. Considering this, the new arrival rate for the new starting event was calculated, in minutes (m), as presented in Table 33.

Table 33 Arrival rate for the four redesign options

Starting event	Volume (q)	Rate (%)	Arrival rate (m)
Incident reported	6035	100%	13.081

Table 34 Impact of R1 implementation

Impact in the 1st SL	Average processing time	
	(m)	(h)
Current state	9.45	0.16
Option R1	8.41	0.14
Difference	1.03	0.02

Table 35 Impact of R2 implementation

Impact	Average processing time					
	Current state		Option R2		Difference	
	(m)	(h)	(m)	(h)	(m)	(h)
1st support level	9.45	0.16	19.50	0.32	10.05	0.16
2nd support level	202.69	3.38	178.17	2.97	24.52	0.41

The redesign options R3 and R4 require the automation of the user activities incident identification and incident categorization in 1st SL. This meant that these two activities would be performed by the IM system, no longer requiring effort for their accomplishment, as already happens with incident logging and incident prioritization. Therefore, the processing time considered for incident identification and incident categorization, in the options R3 and R4, was null minutes.

Options R2, R3 and R4, by addressing issue B, have as consequence the end of dispatching errors. Therefore, the rate and volume concerning incident returned (from the 2nd SL to the 1st SL) were removed in the configuring of these options.

Apart from these nuances in the redesign options, which are due to the changes proposed by each one, the remainder configurations were equal to the ones of Process Analysis.

For each redesign option, an analysis was performed to assess the impact of each one in the time performance of the IM process and on the issues each one addressed.

Option R1 resulted in the reduction of the average processing time for the 1st SL, as shown in Table 34, in minutes (m) and hours (h).

With the employment of H1, issue A is eliminated, which also impacts issue C, resulting in a 10.9% decrease of processing time in the 1st SL when compared with the IM process in its current state. For the worst-case scenario in 1st SL, the results of R1 reveal a decrease to 12.93 min. However, the remainder of the issues still persisted (issue B and issue C), since there was no change in the 2nd SL.

Table 35 presents, in minutes (m) and hours (h), the results of the simulation of option R2.

Although R2 solved issues A, it aggravated issue C by adding 10.05 min to the average processing time in the 1st SL, which means that the time saved by eliminating calls was not enough to compensate for performing the initial diagnosis. For the worst-case scenario in 1st SL, the results of R2 show a decrease to 30.41 min. However, by solving issues B and E, the average processing time for the 2nd SL

Table 36 Impact of R3 implementation

Impact	Average processing time					
	Current state		Option R3		Difference	
	(m)	(h)	(m)	(h)	(m)	(h)
1st support level	9.45	0.16	0	0	9.45	0.16
2nd support level	202.69	3.38	180.98	3.02	21.71	0.36

decreased by 24.52 min, a reduction of 12.1% when compared with the current time performance.

Option R3 produced more significant results, as Table 36 presents in minutes (m) and hours (h).

With the employment of H1, H3 and H4, the option R3 eliminated the processing time for the 1st SL, which solves issues A, B, C and D, and also eliminates the worst-case scenario.

Without dispatching errors, the 2nd SL had it processing time reduced to an average 180.98 min, a break of 10.7% compared to the current state of the IM process. However, in this redesign option, there still was waste of time with invalid incidents (issue E).

The results of option R4, are presented in Table 37, in minutes (m) and hours (h).

By employing all the selected redesign heuristics, R4 solved issue A, B, D and E. The average processing time for 2nd SL was reduced to 178.17 min, a reduction of 12.1% compared to the current time performance. For the worst-case scenario in 1st SL, the results of R4 show a reduction to 30.41 min. However, by deploying all the redesign heuristics, issue C was slightly aggravated, with the processing time for 1st SL increasing to an average 11.5 min, which represents a difference of 2.05 min per incident.

As expected, all the redesign options changed the processing times either for 1st SL and for 2nd SL, and neither altered the processing time for 3rd SL.

Table 38 compares the impact of the four redesign options, in minutes (m) and hours (h).

With the results of the several redesign options analysed, it was clear that R3 and R4 were the redesign options which presented the best time performance improvement.

Table 39 presents a detailed comparison between these two redesign options, in minutes (m) and hours (h).

Table 37 Impact of R4 implementation

Impact	Average processing time					
	Current state		Option R4		Difference	
	(m)	(h)	(m)	(h)	(m)	(h)
1st support level	9.45	0.16	11.5	0.19	2.05	0.03
2nd support level	202.69	3.38	178.17	2.97	24.52	0.41

Table 38 Impact of R4 implementation

Impact	Average processing time							
	Option R1		Option R2		Option R3		Option R4	
	(m)	(h)	(m)	(m)	(m)	(h)	(m)	(h)
1st support level	8.41	0.14	19.50	0.32	0	0	11.50	0.19
2nd support level	202.69	3.38	178.17	2.97	180.98	3.02	178.17	2.97
3rd support level	707.77	11.80	707.77	11.80	707.77	11.80	707.77	11.80

Table 39 Redesign options comparison

Impact	Average processing time					
	Option R3		Option R4		Difference	
	(m)	(h)	(m)	(h)	(m)	(h)
1st support level	0	0	11.50	0.19	11.50	0.19
2nd support level	180.98	3.02	178.17	2.97	2.81	0.05
3rd support level	707.77	11.80	707.77	11.80	–	–

The greatest difference between the two options is the average processing time for the 1st SL, which is 11.50 min. While in R3 there is no time spent with 1st SL activities, resulting from the automation of the dispatching activities, in R4 the processing time is 11.50 min, which is higher than the current time performance. Concerning 2nd SL, the difference between the two options is smaller, consisting only in an average 2.81 min per incident. Once again, all these values are averaged, each hides the reality of possible more extreme, lower or higher, processing times. The two redesign options, R3 and R4, and the respective results were submitted for debate and approval in the 3rd focus group.

The 3rd focus group gathered all the team members available (I17 was no longer part of the team) to debate the redesign options R3 and R4 and its results. The redesign option chosen and approved by the team members was R3. This choice was made by all the team members in consensus, which agreed that R3 was the redesign option that met the issues reported in the 3rd round of interviews and the improvement proposals made in the 4th round of interviews. R4 was rejected due to the higher processing time it demands for the 1st SL, the main source of issues. The team members preferred to keep the initial diagnosis in the 2nd SL and deal with invalid incidents, instead of raising the processing time of 1st SL and receiving only valid, real work incidents in the 2nd SL. Also, management rules from the organisation required that the initial diagnosis to remain in the 2nd SL, due to an update of the IM system that was planned to occur, has a step for the improvement of the IM process. The final approved to-be model for the IM process is depicted below in Fig. 5.

The 3rd focus group was the finishing step of Process Redesign.

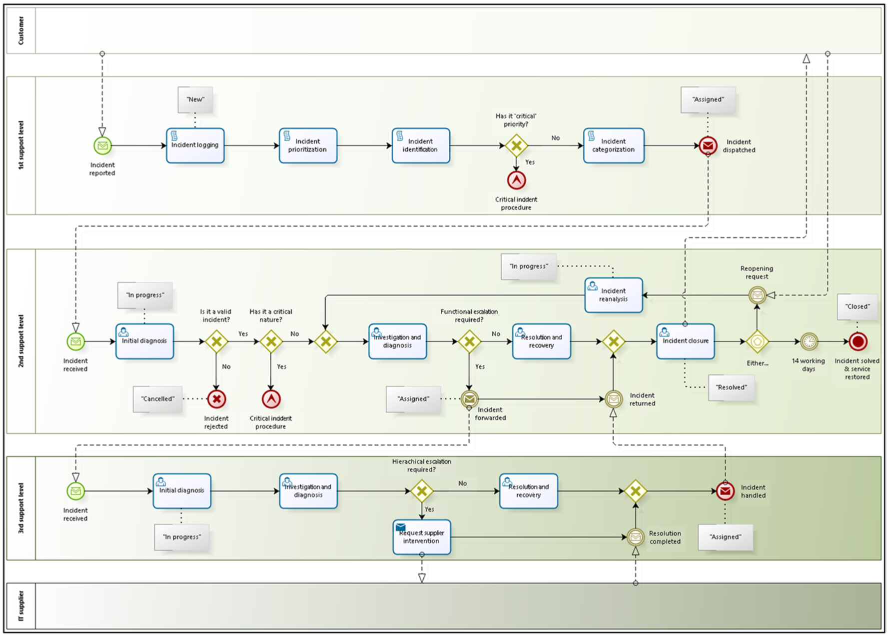


Fig. 5 To-Be model

7 Lessons learned and recommendations

The conduction of the case study revealed how the analysed IM process is shaped, what is its current performance and how that performance can be improved through the employment of the BPM methodology and redesign heuristics.

The initial phase of the case study, Process Discovery, provided both a visual and a documented understanding of how the IM process is performed and who are the participants involved. It was discovered that the analysed IM process is a standardized process, mainly based on the ITIL standard. The IM process not only follows the ITIL workflow, but also most of the recommendations and rules proposed by ITIL for the management of incidents' lifecycle are adopted by the organisation's support service. The results and conclusions of this initial phase, namely the as-is model, were the basis for the analysis performed in the ensuing phase.

The second phase of the case study, Process Analysis, allowed to assess, characterize and quantify the IM process in its current state, and to reveal and measure the issues affecting its time performance. A list of five identified and quantified issues affecting the IM process current time performance was produced and was assessed and approved by the team members. Of those five issues, four had an operational nature, while one had a technological nature, being the issues mainly related to the 1st SL. These identified issues were targeted as improvement opportunities for the final phase.

Process Redesign, the last phase in the conduction of the case study, analysed how the issues identified in Process Analysis could be resolved and the IM process improved, through the identification and employment of BPM heuristics, which consists in researching goal of the case study. Four redesign heuristics (activity elimination, resequencing, activity automation and integral technology) were identified has best suited for employment in the IM process to overcome its issues and improve its time performance. Based in the selected heuristics, four different redesign options were modelled and simulated in the BPM system. Option R3 was chosen by the team as the to-be version of the IM process, mainly because it addressed the issues reported by the team members in the 3rd round of interviews and reflected the improvement proposals suggested in the 4th round of interviews. It was also chosen due to management reasons, which had a planned upgrade of the used IM system that was starting to happen. Thus, the applied redesign heuristics for improvement of the IM process were activity elimination, activity automation and integral technology.

Comparing the as-is with the to-be model, the employment of activity elimination, activity automation and integral technology enabled to address and solve the issues identified during Process Analysis. The elimination of calls as entry point in the IM process allowed to remove a time-consuming activity and streamlined the arrival of incidents. This in turn enabled the full automation of the 1st SL activities, by upgrading the IM system. The measured result was the elimination of the time spent by the 1st SL and the decrease of 10.7% in the average processing time required by the 2nd SL, when compared with the current state of the IM process. Thus, the IM process become faster, which was the desired outcome.

As managerial recommendations, this case study reveals that the upgrading of the IM system is a good option and required for the automation of activities to be possible. The automation of activities solves the waste of time with dispatching activities and with dispatching errors, existing in the current situation. However, it is fundamental to eliminate calls and establish emails as the single-entry point of incidents, not only because it saves time, but also because it allows the automation of activities to achieve its full effect. With the automation of all the activities of the 1st SL, there is no need for HR in this SL. These team members can be trained to perform 2nd SL role, increasing its installed capacity. The establishment of KPIs to measure effort and processing times allows to control the time performance of the IM process and to discover and approach potential issues more easily. The implementation of such parameters is recommended while updating the IM system.

8 Discussion

This case study shows that, a good process can always be improved. Although the analysed IM process was complying with its operational and quality targets, there was still room for improvement of the time performance. The identified and selected redesign heuristics—activity automation, activity elimination and integral technology—proved to be the best suited to be employed for the improvement of the analysed IM process, while respecting the ITIL guidelines.

As seen in Sect. 2, to the best of our knowledge, there is no related work found applying a BPM approach for the improvement of the IM process specifically.

In the literature review found concerning the improvement of the IM process, seen in, researches study and propose solutions for the improvement and optimization of the IM process, being automation a frequently explored path. This further sustains the results of this case study on indicating activity automation and integral technology as best suited redesign heuristics. Different techniques and methods are presented for automating IM process. This may be due to the ease existent for the automation of certain activities in the IM process, especially the less analytical and time-consuming ones. Such is the case in this research, as the activities performed by the 1st SL are proposed for automation. Other solutions for improvement are presented. However, these solutions are IT-oriented, looking at improvement as a technical challenge and focusing on specific features and activities of the IM process, not considering the whole processual perspective. Contrary to this, the BPM approach applied in this research allows organisations to have an integrated perspective of the IM process, understanding the relations between the different activities and participants, and providing an end-to-end view of how a given improvement can affect the performance the IM process.

9 Conclusion

BPM is a broad discipline that offers methodologies and tools for the control and improvement of BP. By using BPM, managers can thoroughly analyse BP and discover improvement opportunities, which is increasingly requested in the organisational environment.

This research explored how the IM process can be improved through a BPM approach, a relationship that has not yet been much investigated by the scientific community. For this goal, a case study methodology was followed. The conducted case study followed the recommendations suggested by Yin for the validation of the research, thus assuring the quality of the results obtained, as Table 40 describes.

The goal of the case study conducted was achieved with the presentation of the BPM redesign heuristics that best suit the analysed IM process. This means that the objective of the research was also achieved, by showing that the IM process can be improved through a BPM approach.

The main conclusions of this research are that the best suited redesign heuristics for the improvement of the analysed IM process are activity elimination, activity automation and integral technology. IT managers and organisations should consider these as a way to achieve a better performance for the IM process.

This research contributes to reduce the gap existent between BPM and ITSM processes. There is a clear relation between both areas, due to their process-oriented nature, but there is very few researches developed in this specific area.

Although the research fulfilled its objectives, it faced some limitations. First, the main one, this research itself is limited to this organisation, to this case study, which means that results obtained may not be valid for other cases. However, this kind of process is highly adopted by organisations and also is heavily standardized, being ITIL the most adapted standard. This means that, although this is a case study

Table 40 Validity tests to case study methodology

Test	Description	Sections
Construct validity	Multiple sources of evidence are used, and triangulation of data is also performed	3, 4 and 5
Internal validity	Causal relationship is assured, by producing explanations and conclusions through the analysis and linking of the collected evidence	4 and 5
External validity	A literature review is performed to define the domain to which the study's findings can be generalized. The results of the research are compared with the related work found	2, 6 and 7
Reliability	The procedures of the case study are documented and Yin (2009) recommendations were followed	3

limited to the chosen unit of analysis, its results may be similarly adopted for other cases. The limitations found in the collection of evidence, concerning the archival records, did not allowed a more solid, in-depth analysis to the IM process, which could have revealed more issues.

For future researches on the topic, BPM is a management approach that can help to understand and improve IT processes. Researchers and managers can start looking at IT processes as common industrial operations that can be improved to assure the best service provided. This relationship can be more explored, in larger scopes, for example, the analysis and improvement of all processes in a support service. As mentioned, there is a large gap to close in this area. Still, it is advised the study and knowledge of BPM analytical tools and of ITSM frameworks to perform this kind of research.

References

AXELOS. (2019). *ITIL Foundation: ITIL 4 Edition*. TSO (The Stationery Office).

Bartolini C, Stefanelli C, Tortonesi M (2008) SYMIAN: a simulation tool for the optimization of the IT incident management process. In: Turck F, Kellerer W, Kormentzas G (eds) 19th IFIP/IEEE international workshop on distributed systems: operations and management. IEEE, Samos Island, Greece, pp 83–94

Becker J, Fischer R, Janiesch C (2007) Optimizing US healthcare processes: a case study in business process management. In: 13th Americas conference on information systems, vol 4. Keystone, CO, USA, pp 2236–2247.

Bezerra G, Pinheiro V, Bessa A (2014) Incident management optimization through the reuse of experiences and natural language processing. In: da Silva AR, Silva AR, Brito MA, Machado RJ (eds) 9th international conference on the quality of information and communications technology. IEEE, Guimarães, Portugal, pp 58–65

BMC, Forbes Insights (2017) Delivering value to today's digital enterprise: the state of IT service management. In: Moreno KW, Moreno HS, McKendrick J, Crean KS (eds). Forbes Insights, Jersey City, NJ, USA.

Conger S, Winniford M, Erickson-Harris L (2008) Service management in operations. 14th Americas conference on information systems. Toronto, ON, Canada, AIS, pp 3884–3894

de Morais RM, Kazan S, de Pádua SID, Costa AL (2014) An analysis of BPM lifecycles: from a literature review to a framework proposal. Bus Process Manag J 20(3):412–432. https://doi.org/10.1108/BPMJ-03-2013-0035

Dumas M, Rosa ML, Mendling J, Reijers HA (2013) Fundamentals of business process management. Springer Publishing Company, Incorporated

Ferretti M, Schiavone F (2016) Internet of things and business processes redesign in seaports: the case of Hamburg. Bus Process Manag J 22(2):271–284. https://doi.org/10.1108/BPMJ-05-2015-0079

Galup SD, Dattero R, Quan JJ, Conger S (2009) An overview of IT service management. Commun ACM 52(5):124. https://doi.org/10.1145/1506409.1506439

Ghrab I, Ketata M, Loukil Z, Gargouri F (2016) Using constraint programming techniques to improve incident management process in ITIL. In: 2016 Third International Conference on Artificial Intelligence and Pattern Recognition (AIPR), pp 1–6. IEEE. https://doi.org/https://doi.org/10.1109/ICAIPR.2016.7585231

Goby N, Brandt T, Feuerriegel S, Neumann D (2016) Business intelligence for business processes: the case of IT incident management. 24th European conference on information systems. Istanbul, Turkey, AIS, pp 1–15

Gupta R, Prasad KH, Mohania M (2008a) Automating ITSM incident management process. 5th international conference on autonomic computing, ICAC 2008, vol 1, pp 141–150. https://doi.org/https://doi.org/10.1109/ICAC.2008.22

Gupta R, Prasad KH, Mohania M (2008b) Information integration techniques to automate incident management. In: NOMS 2008 - IEEE/IFIP Network Operations and Management Symposium: Pervasive Management for Ubiquitous Networks and Services, pp 979–982. https://doi.org/https://doi.org/10.1109/NOMS.2008.4575262

Haddad CR, Ayala DHF, Maldonado MU, Forcellini FA, Lezana ÁGR (2016) Process improvement for professionalizing non-profit organizations: BPM approach. Business Proc Manag J 22(3):634–658. https://doi.org/10.1108/MRR-09-2015-0216

Hanafizadeh P, Moosakhani M, Bakhshi J (2009) Selecting the best strategic practices for business process redesign. Bus Process Manag J 15(4):609–627. https://doi.org/10.1108/14637150910975561

Harmon P (2010) The scope and evolution of business process management. Handbook on Business Process Management, vol 1, pp 37–81. https://doi.org/https://doi.org/10.1007/978-3-642-00416-2

Harmon P (2014) Business process change, 3rd edition, vol 53. Elsevier B.V., Waltham, MA, USA. https://doi.org/https://doi.org/10.1017/CBO9781107415324.004

Hertz S, Johansson JK, de Jager F (2001) Customer-oriented cost cutting: process management at Volvo. Supp Chain Manag Int J 6(3):128–141. https://doi.org/10.1108/13598540110399174

Iden J, Eikebrokk TR (2013) Implementing IT service management: a systematic literature review. Int J Inf Manage 33(3):512–523. https://doi.org/10.1016/j.ijinfomgt.2013.01.004

Jamous, N., Bosse, S., Gorling, C., Hintsch, J., Khan, A., Kramer, F., … Turowski, K. (2017). Towards an IT Service Lifecycle Management (ITSLM) Concept. In *Proceedings - 4th International Conference on Enterprise Systems: Advances in Enterprise Systems, ES 2016* (pp. 29–38). https://doi.org/https://doi.org/10.1109/ES.2016.10

Küng P, Hagen C (2007) The fruits of business process management: an experience report from a swiss bank. Bus Process Manag J 13(4):477–487. https://doi.org/10.1108/14637150710763522

Mahy Y, Ouzzif M, Bouragba K (2016) Supporting ITIL processes implementation using business process management systems. In: Ouzzif M, Jarir Z (eds) 3rd international conference on systems of collaboration. IEEE, Casablanca, MO, pp 77–80

Mansar SL, Reijers HA (2007) Best practices in business process redesign: use and impact. Bus Process Manag J 13(2):193–213. https://doi.org/10.1108/14637150710740455

Marrone M, Kolbe LM (2011) Impact of IT service management frameworks on the IT organization. Bus Inf Syst Eng 3(1):5–18. https://doi.org/10.1007/s12599-010-0141-5

Mohamed N, Kaur J, Singh G (2012) A conceptual framework for information technology governance effectiveness in private organizations. Inf Manag Comput Secur 20(2):88–106

Netjes M, Mans RS, Reijers HA, Van Der Aalst WMP, Vanwersch RJB (2010) BPR Best practices for the healthcare domain. In: Rinderle-Ma S, Sadiq S, Leymann F (eds) Workshops of 7th international conference on business process management, vol 43. Springer, Berlin, pp 605–616

Pereira R, Mira da Silva M (2012) Designing a new integrated IT governance and IT management framework based on both scientific and practitioner viewpoint. Int J Enterp Inf Syst 8(4):1–43. https://doi.org/10.4018/jeis.2012100101

Pereira R, Velez Lapão L, Scalabrin Bianchi I, Amaral D (2020) Improving emergency department through business process redesign: an empirical study. Australas J Inf Syst. https://doi.org/10.3127/ajis.v24i0.2679

Porter M (1980) Competitive strategy: techniques for analyzing industries and competitors, 1st edn. Free Press, New York, NY, USA

Pyon CU, Woo JY, Park SC (2011) Service improvement by business process management using customer complaints in financial service industry. Expert Syst Appl 38(4):3267–3279. https://doi.org/10.1016/j.eswa.2010.08.112

Rebuge Á, Ferreira D (2012) Business process analysis in healthcare environments: a methodology based on process mining. Journal of Information Systems, vol 37.Springer, Berlin.

Rinaldi M, Montanari R, Bottani E (2015) Improving the efficiency of public administrations through business process reengineering and simulation: a case study. Bus Process Manag J 21(2):419–462. https://doi.org/10.1108/BPMJ-06-2014-0054

Roeser T, Kern E-M (2015) Surveys in business process management - a literature review. Bus Process Manag J 21(3):692–718. https://doi.org/10.1108/BPMJ-07-2014-0065

Runeson P, Höst M (2009) Guidelines for conducting and reporting case study research in software engineering. Empir Softw Eng 14(2):131–164. https://doi.org/10.1007/s10664-008-9102-8

Salah S, Maciá-Fernández G, Díaz-Verdejo JE, Sánchez-Casado L (2015) A model for incident tickets correlation in network management. J Netw Syst Manage 24(1):57–91. https://doi.org/10.1007/s10922-014-9340-6

Sallé M (2004) IT service management and IT governance: review, comparative analysis and their impact on utility computing. Hewlett-Packard Company.
Sanka Laar D, Seymour LF (2017) Redesigning business processes for small and medium enterprises in developing countries. In: Mokoaleli-Mokoteli T , Ndaba Z (eds). 13th European Conference on management, leadership & governance, pp. 512–519. Academic Conferences and Publishing International Ltd, Saint Petersburg, Russia.
Santos F, Pereira R, Vasconcelos JB (2019) Toward robotic process automation implementation: an end-to-end perspective. Bus Process Manag J. https://doi.org/10.1108/BPMJ-12-2018-0380
Valiente M-C, Garcia-Barriocanal E, Sicilia M-A (2012) Applying an ontology approach to IT service management for business-IT integration. Knowl-Based Syst 28:76–87. https://doi.org/10.1016/j.knosys.2011.12.003
van der Aalst WMP (2013) Business process management: a comprehensive survey. ISRN Software Eng 2013:1–37. https://doi.org/10.1155/2013/507984
Weske M (2012) Business Process management. Business process management: concepts, languages, architectures, 2nd edn. Springer, Berlin Heidelberg, pp 3–23
Yin R (2009) Case study research and applications: design and methods, 5th edn. SAGE Publications, Thousand Oaks, CA, USA
Zaidah, Z. (2007). Case study as a research method. In: Jurnal Kemanusiaan, vol. 9, pp. 1–6. https://doi.org/https://doi.org/10.1177/15222302004003007
Zainal Z (2007) Case study as a research method. Jurnal Kemanusiaan. 9:1–6. https://doi.org/10.1177/15222302004003007

Rúben Pereira is an assistant professor at ISCTE. He has a PhD in information systems at IST where he also graduated as a master in computer engineering. He is the director of the master in computer science and business management. He is the author of several scientific papers in the area of IT Governance.

José Braga de Vasconcelos has a PhD in Computer Science at the University of York (UK). Research in the area of Knowledge Engineering, including several publications on relevant journals. Lecturer of algorithms and data structures and software engineering at the Universidade Europeia in Lisbon, and at the Universidade Lusófona in Porto.

Álvaro Rocha is a professor of Information Systems at ISEG, University of Lisbon, President of AISTI (Iberian Association for IS), Chair of the IEEE Portugal Section Systems, Man, and Cybernetics Society Chapter, and Editor-in-Chief of both JISEM and RISTI.

Isaías Scalabrin Bianchi has a Ph.D. in Information Systems at the University of Minho, Portugal. He is a civil servant at Federal University of Santa Catarina. He also collaborates at the Al-Farabi Kazakh National University.